INDEPENDENCE OF THE PRESENTATION IN THE CONSTRUCTION OF THE TWISTED INVERSE IMAGE FUNCTOR

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The identification of the functor attached to a presentation as the right adjoint of the restriction along it. Invoked inside the counted proof (source lines 4612-4615). Makes the functor for the composed presentation the composite of the two functors.
- The computation of the functor for the extension of a presentation by a single variable. Invoked at the end of the counted proof (source lines 4624-4625). Identifies the single step with the shift of the tensor product.

Source reference. Original lemma lemma-compute-for-effective-Cartier-algebraic: sources/stacks-dualizing/pinned-original-source.tex, source lines 2621–2621. Navigation pointer only: the counted text refers to the Stacks item lemma-compute-for-effective-Cartier-algebraic, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original section section-trivial: sources/stacks-dualizing/pinned-original-source.tex, source lines 2416–2416. Navigation pointer only: the counted text refers to the Stacks item section-trivial, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $\varphi : R \to A$ be a finite type homomorphism of Noetherian rings. The functor $\varphi^!$ is well defined up to isomorphism.*

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Proof. Suppose that $\psi_1 : P_1 = R[x_1, \ldots, x_n] \to A$ and $\psi_2 : P_2 = R[y_1, \ldots, y_m] \to A$ are two surjections from polynomial rings onto A. Then we get a commutative diagram

$$\begin{array}{ccc} R[x_1, \ldots, x_n, y_1, \ldots, y_m] & \xrightarrow[y_j \mapsto f_j]{} & R[x_1, \ldots, x_n] \\ \downarrow{\scriptstyle x_i \mapsto g_i} & & \downarrow \\ R[y_1, \ldots, y_m] & \longrightarrow & A \end{array}$$

where f_j and g_i are chosen such that $\psi_1(f_j) = \psi_2(y_j)$ and $\psi_2(g_i) = \psi_1(x_i)$. By symmetry it suffices to prove the functors defined using $P \to A$ and $P[y_1, \ldots, y_m] \to A$ are isomorphic. By induction we may assume $m = 1$. This reduces us to the case discussed in the next paragraph.

Here $\psi : P \to A$ is given and $\chi : P[y] \to A$ induces ψ on P. Write $Q = P[y]$. Choose $g \in P$ with $\psi(g) = \chi(y)$. Denote $\pi : Q \to P$ the P-algebra map with $\pi(y) = g$. Then $\chi = \psi \circ \pi$ and hence $\chi^! = \psi^! \circ \pi^!$ as both are adjoint to the restriction functor $D(A) \to D(Q)$ by the material in Section section-trivial. Thus

$$\chi^! \left(K \otimes^{\mathbf{L}}_R Q\right)[n+1] = \psi^! \left(\pi^! \left(K \otimes^{\mathbf{L}}_R Q\right)[1]\right)[n]$$

Hence it suffices to show that $\pi^!(K \otimes^{\mathbf{L}}_R Q[1]) = K \otimes^{\mathbf{L}}_R P$ Thus it suffices to show that the functor $\pi^!(-) : D(Q) \to D(P)$ is isomorphic to $K \mapsto K \otimes^{\mathbf{L}}_Q P[-1]$. This follows from Lemma lemma-compute-for-effective-Cartier-algebraic. □